

INITIAL EMPLOYEE DATA

ACID_Test_hr_demo x

Limit to 1000 rows

```
37
38 -- =====
39 -- 4. VIEW INITIAL DATA
40 -- =====
41 • SELECT 'INITIAL EMPLOYEE DATA' AS info;
42 • SELECT * FROM employees;
43
```

Result Grid | Filter Rows: | Edit: | Export/Import: |

	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	IT	800000.00	10
	2	Su Su	HR	600000.00	8
	3	Kyaw Kyaw	Finance	900000.00	12
	4	Hla Hla	Admin	550000.00	6
	5	Mya Mya	IT	750000.00	9
	6	Ko Ko	Operations	700000.00	11
*	NULL	NULL	NULL	NULL	NULL

ATOMICITY TEST - BEFORE SUCCESSFUL TRANSACTION

ACID_Test_hr_demo x

Limit to 1000 rows

```
54
55 • SELECT 'ATOMICITY TEST - BEFORE SUCCESSFUL TRANSACTION' AS info;
56 • SELECT * FROM employees WHERE employee_id = 1;
57
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	IT	800000.00	10
*	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET salary = salary + 50000

WHERE employee_id = 1;

```

UPDATE employees
SET leave_balance = leave_balance - 2
WHERE employee_id = 1;

COMMIT;

```

ATOMICITY TEST - AFTER SUCCESSFUL COMMIT

ACID_Test_hr_demo x

Limit to 1000 rows

```

73
74 • SELECT 'ATOMICITY TEST - AFTER SUCCESSFUL COMMIT' AS info;
75 • SELECT * FROM employees WHERE employee_id = 1;
76

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content

	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	IT	850000.00	8
*	NULL	NULL	NULL	NULL	NULL

ATOMICITY TEST - BEFORE FAILED TRANSACTION

ACID_Test_hr_demo x

Limit to 1000 rows

```

81 -----
82 • SELECT 'ATOMICITY TEST - BEFORE FAILED TRANSACTION' AS info;
83 • SELECT * FROM employees WHERE employee_id = 2;
84

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content

	employee_id	employee_name	department	salary	leave_balance
▶	2	Su Su	HR	600000.00	8
*	NULL	NULL	NULL	NULL	NULL

```

START TRANSACTION;

```

```

UPDATE employees
SET salary = salary + 50000

```

```
WHERE employee_id = 2;
```

```
-- Mistake / unexpected issue
```

```
-- This update affects no valid intended record
```

```
UPDATE employees
```

```
SET leave_balance = leave_balance - 100
```

```
WHERE employee_id = 999;
```

```
ROLLBACK;
```

ATOMICITY TEST - AFTER ROLLBACK

ACID_Test_hr_demo x

98

99 • SELECT 'ATOMICITY TEST - AFTER ROLLBACK' AS info;

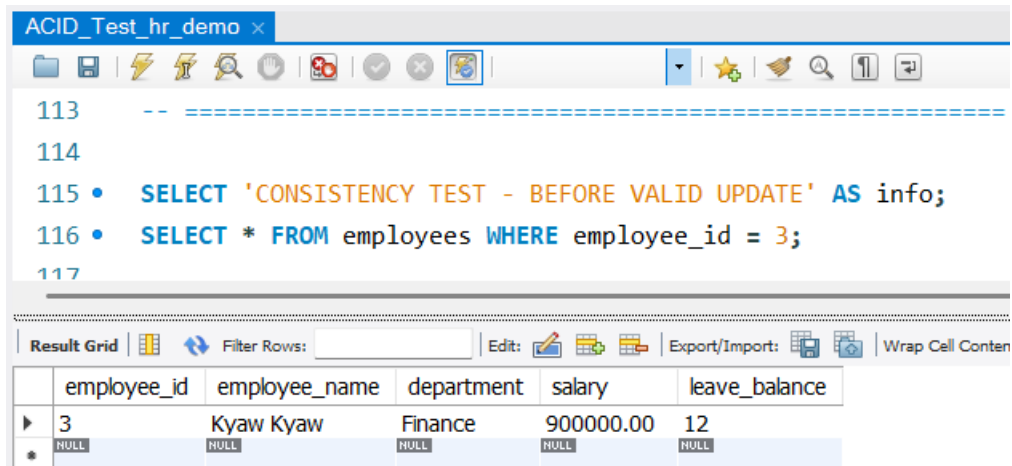
100 • SELECT * FROM employees WHERE employee_id = 2;

101

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

employee_id	employee_name	department	salary	leave_balance
2	Su Su	HR	600000.00	8
*	NULL	NULL	NULL	NULL

CONSISTENCY TEST - BEFORE VALID UPDATE



The screenshot shows a SQL Developer window titled "ACID_Test_hr_demo". The query window contains the following SQL code:

```
113  -- =====
114
115 • SELECT 'CONSISTENCY TEST - BEFORE VALID UPDATE' AS info;
116 • SELECT * FROM employees WHERE employee_id = 3;
117
```

The result grid below the query window displays the following data:

employee_id	employee_name	department	salary	leave_balance
3	Kyaw Kyaw	Finance	900000.00	12

START TRANSACTION;

UPDATE employees

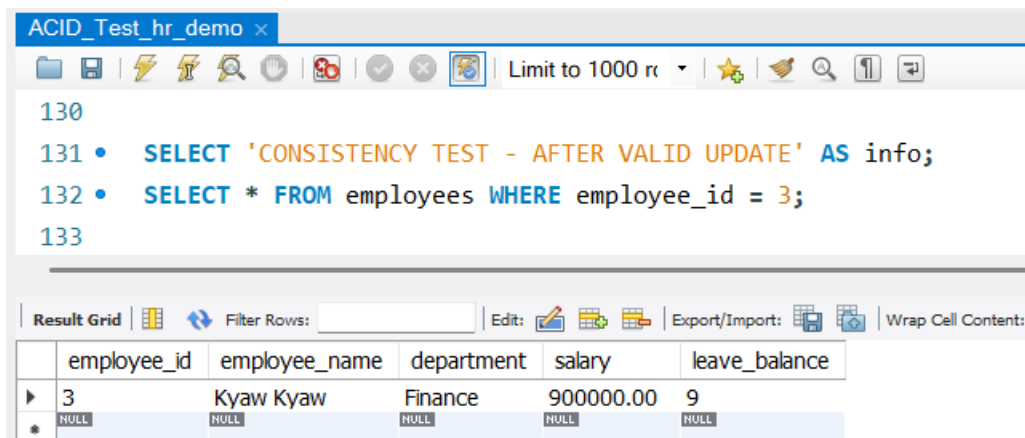
SET leave_balance = leave_balance - 3

WHERE employee_id = 3

AND leave_balance >= 3;

COMMIT;

CONSISTENCY TEST - AFTER VALID UPDATE



The screenshot shows the same SQL Developer window after the update. The query window contains the following SQL code:

```
130
131 • SELECT 'CONSISTENCY TEST - AFTER VALID UPDATE' AS info;
132 • SELECT * FROM employees WHERE employee_id = 3;
133
```

The result grid below the query window displays the following data:

employee_id	employee_name	department	salary	leave_balance
3	Kyaw Kyaw	Finance	900000.00	9

CONSISTENCY TEST - BEFORE INVALID ATTEMPT

ACID_Test_hr_demo x

Limit to 1000 r

```
137
138 • SELECT 'CONSISTENCY TEST - BEFORE INVALID ATTEMPT' AS info;
139 • SELECT * FROM employees WHERE employee_id = 4;
140
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	employee_id	employee_name	department	salary	leave_balance
▶	4	Hla Hla	Admin	550000.00	6
*	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET leave_balance = leave_balance - 20

WHERE employee_id = 4

AND leave_balance >= 20;

COMMIT;

CONSISTENCY TEST - AFTER INVALID ATTEMPT (NO CHANGE EXPECTED)

ACID_Test_hr_demo x

Limit to 1000 r

```
149
150 • SELECT 'CONSISTENCY TEST - AFTER INVALID ATTEMPT (NO CHANGE EXPECTED)' AS info;
151 • SELECT * FROM employees WHERE employee_id = 4;
152
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	employee_id	employee_name	department	salary	leave_balance
▶	4	Hla Hla	Admin	550000.00	6
*	NULL	NULL	NULL	NULL	NULL

ISOLATION TEST

ACID_Test_hr_demo x

Limit to 1000 r

```
170 -- Run the following in Window 1:
171 --
172 • USE hr_demo;
173 • SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
174 • START TRANSACTION;
175 --
176 • UPDATE employees
177   SET salary = salary + 100000
178   WHERE employee_id = 5;
179 --
180 • SELECT * FROM employees WHERE employee_id = 5;
```

Result Grid

	employee_id	employee_name	department	salary	leave_balance
▶	5	Mya Mya	IT	850000.00	9
*	NULL	NULL	NULL	NULL	NULL

ACID_IsolationTest_hr_d... x

Limit to 1000 r

```
1 -- SESSION 2
2 -- -----
3 -- Run the following in Window 2:
4 --
5 • USE hr_demo;
6 • SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
7 --
8 • SELECT * FROM employees WHERE employee_id = 5;
9 --
10 -- Expected:
11 -- Session 2 should still see the old committed salary.
```

Result Grid

	employee_id	employee_name	department	salary	leave_balance
▶	5	Mya Mya	IT	750000.00	9
*	NULL	NULL	NULL	NULL	NULL

-- Then go back to Session 1 and run:

COMMIT;

ACID_IsolationTest_hr_d... x

Limit to 1000 r

```

1  -- SESSION 2
2  -- -----
3  -- Run the following in Window 2:
4  --
5  • USE hr_demo;
6  • SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
7  --
8  • SELECT * FROM employees WHERE employee_id = 5;
9  --
10 -- Expected:
11 -- Session 2 should still see the old committed salary.

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	employee_name	department	salary	leave_balance
▶	5	Mya Mya	IT	850000.00	9
*	NULL	NULL	NULL	NULL	NULL

DURABILITY TEST - BEFORE COMMIT

ACID_Test_hr_demo x

Limit to 1000 r

```

218 -- =====
219
220 • SELECT 'DURABILITY TEST - BEFORE COMMIT' AS info;
221 • SELECT * FROM employees WHERE employee_id = 2;
222

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

	employee_id	employee_name	department	salary	leave_balance
▶	2	Su Su	HR	600000.00	8
*	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET salary = 950000

WHERE employee_id = 2;

COMMIT;

DURABILITY TEST - AFTER COMMIT

ACID_Test_hr_demo x

Limit to 1000 rows

```
230
231 • SELECT 'DURABILITY TEST - AFTER COMMIT' AS info;
232 • SELECT * FROM employees WHERE employee_id = 2;
233
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	employee_id	employee_name	department	salary	leave_balance
▶	2	Su Su	HR	950000.00	8
*	NULL	NULL	NULL	NULL	NULL

PROMOTION TEST - BEFORE SUCCESSFUL PROMOTION

ACID_Test_hr_demo x

Limit to 1000 rows

```
253
254 • SELECT 'PROMOTION TEST - BEFORE SUCCESSFUL PROMOTION' AS info;
255 • SELECT * FROM employees WHERE employee_id = 1;
256
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	IT	850000.00	8
*	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET salary = salary + 150000

WHERE employee_id = 1;

UPDATE employees

SET department = 'Senior IT'


```
WHERE employee_id = 1;
```

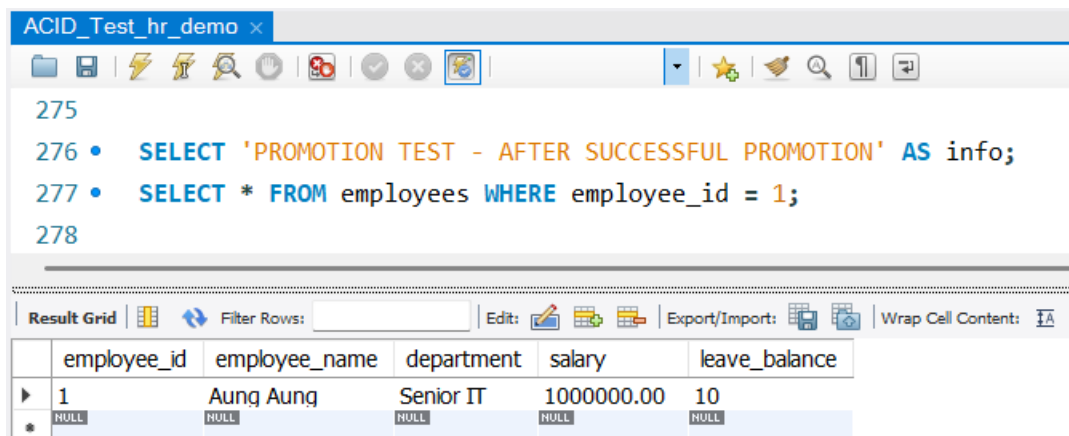
```
UPDATE employees
```

```
SET leave_balance = leave_balance + 2
```

```
WHERE employee_id = 1;
```

```
COMMIT;
```

PROMOTION TEST - AFTER SUCCESSFUL PROMOTION



ACID_Test_hr_demo x

275

276 • **SELECT** 'PROMOTION TEST - AFTER SUCCESSFUL PROMOTION' **AS** info;

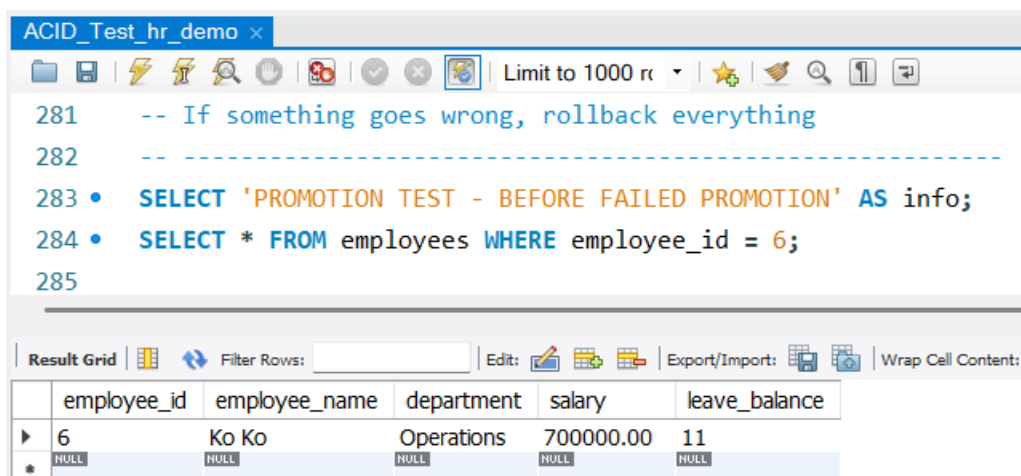
277 • **SELECT** * **FROM** employees **WHERE** employee_id = 1;

278

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	Senior IT	1000000.00	10
*	NULL	NULL	NULL	NULL	NULL

PROMOTION TEST - BEFORE FAILED PROMOTION



ACID_Test_hr_demo x

281 -- If something goes wrong, rollback everything

282 -----

283 • **SELECT** 'PROMOTION TEST - BEFORE FAILED PROMOTION' **AS** info;

284 • **SELECT** * **FROM** employees **WHERE** employee_id = 6;

285

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	employee_name	department	salary	leave_balance
▶	6	Ko Ko	Operations	700000.00	11
*	NULL	NULL	NULL	NULL	NULL

```
START TRANSACTION;
```

```
UPDATE employees
```

```
SET salary = salary + 150000
```

```
WHERE employee_id = 6;
```

```
UPDATE employees
```

```
SET department = 'Senior Operations'
```

```
WHERE employee_id = 6;
```

```
-- Suppose this is wrong
```

```
UPDATE employees
```

```
SET leave_balance = leave_balance + 2
```

```
WHERE employee_id = 999;
```

```
ROLLBACK;
```

PROMOTION TEST - AFTER FAILED PROMOTION ROLLBACK

The screenshot shows a SQL IDE window titled "ACID_Test_hr_demo". The query window contains the following SQL commands:

```
301 • ROLLBACK;  
302  
303 • SELECT 'PROMOTION TEST - AFTER FAILED PROMOTION ROLLBACK' AS info;  
304 • SELECT * FROM employees WHERE employee_id = 6;  
305
```

Below the query window is a "Result Grid" showing the results of the last query. The grid has columns: employee_id, employee_name, department, salary, and leave_balance. The first row shows the results for employee_id 6.

	employee_id	employee_name	department	salary	leave_balance
▶	6	Ko Ko	Operations	700000.00	11
*	NULL	NULL	NULL	NULL	NULL

FINAL EMPLOYEE DATA

ACID_Test_hr_demo x

Limit to 1000 r

```
305
306  -- =====
307  -- FINAL TEST DATA VIEW
308  -- =====
309 • SELECT 'FINAL EMPLOYEE DATA' AS info;
310 • SELECT * FROM employees;
311
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	Senior IT	1000000.00	10
	2	Su Su	HR	950000.00	8
	3	Kyaw Kyaw	Finance	900000.00	9
	4	Hla Hla	Admin	550000.00	6
	5	Mya Mya	IT	850000.00	9
	6	Ko Ko	Operations	700000.00	11
*	NULL	NULL	NULL	NULL	NULL